ZUZER CHATHIWALA

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PROFESSIONAL SUMMARY

Final-year B. Tech Bioinformatics student (CGPA 8.54, Marwadi University NAAC A+) with proven experience in machine learning, deep learning, and computational biology applied to clinical and genomic datasets. IEEE-published across oncology, neurology, and hematology. Summer Research Fellow at IIEST Shibpur, built PyTorch-based predictive models on physiological signal data. Proficient in Python-based bioinformatics pipelines, multi-omics data analysis, structural bioinformatics (PyMOL, AlphaFold, PDB), and transcriptomics workflows. Ready to contribute AI/ML expertise toward genomics, drug target discovery, and computational biology in a biotech startup environment.

CORE SKILLS

AI / ML / DL: Python, PyTorch, Scikit-learn, NumPy, Pandas, Deep Learning, Neural Networks, Feature Engineering, Predictive Modeling, Signal Processing

Bioinformatics & Genomics: BLAST, MEGA 12, FastQC, Trimmomatic, STRING, KEGG, Ensembl, NCBI, GEO/SRA datasets, Functional Enrichment, Pathway Analysis, Transcriptomics, RNA-Seq, scRNA-seq/Scanpy.

Structural Bioinformatics: PyMOL, PDB format, AlphaFold, Protein–Protein Interactions, AutoDock/AutoDock Vina (exposure), Molecular Docking principles

Bioinformatics Platforms: DESeq2, g:Profiler, Pathview, Cytoscape, DrugBank

Domain Knowledge: Cancer Genomics, Differential Gene Expression, Computational Drug Discovery

RESEARCH EXPERIENCE

Summer Research Fellow — IIEST Shibpur (Indian Institute of Engineering Science & Technology) *May–July 2025*

- Designed and trained PyTorch deep learning models on high-dimensional physiological signal datasets for biomedical pattern recognition.
- Applied signal preprocessing, feature extraction, and model evaluation pipelines directly transferable to large-scale genomic and protein sequence analysis.
- Gained hands-on experience in data-driven neural network modeling and performance benchmarking on biological data.

PROJECTS

Integrated Bioinformatics Analysis of Hepatocellular Carcinoma (Minor Research Project) *2026-2027*

- Identifying prognostic biomarkers and therapeutic targets in liver cancer using TCGA-LIHC dataset via differential gene expression (DESeq2), functional enrichment (g:Profiler), pathway visualization (Pathview), PPI network analysis (STRING, Cytoscape), and drug–gene mapping (DrugBank).

Single-Cell Transcriptomic Analysis: Differential Gene Expression & Cell-Type Annotation April *2026*

- Engineered reproducible, end-to-end Python workflows for scRNA-seq analysis, encompassing quality control, clustering, differential expression, and cell-type annotation of public datasets to investigate disease-associated transcriptional changes and identify candidate biomarkers.

Sleep Apnea Detection & CPAP Therapy Optimization via Deep Learning *2025–2026*

- Built deep learning predictive frameworks (PyTorch) for automated sleep apnea detection from physiological signals; applied feature engineering and signal preprocessing. Accepted: IEEE I3CTCON 2026.

Breast Cancer Outcome Prediction — Integrated ML Classifiers *2024–2025*

- Applied ensemble ML classifiers (Scikit-learn) on oncology datasets for prognostic feature identification and survival outcome prediction. Published: IEEE HTC Japan 2025.

Parkinson's Disease Diagnosis — Hybrid ML Parameter Optimization *2024–2025*

- Designed a hybrid ML pipeline with comparative classifier analysis to maximize diagnostic sensitivity from clinical/voice feature datasets. Published: IEEE HTC Japan 2025.

Alpha Thalassemia Classification via Machine Learning *2024–2025*

- Built ML classification model (Python, Scikit-learn) for hematological disorder variant classification. Accepted for oral presentation: IEEE ASSIC 2025.

PUBLICATIONS

- Sleep Apnea Detection & CPAP Therapy Adjustment via Deep Learning — IEEE I3CTCON 2026 (Accepted)
- Predicting Breast Cancer Outcomes Using Integrated ML Classifiers — IEEE HTC Japan 2025 (Published)
- Optimized Parameter Selection for Parkinson's Diagnosis via Hybrid ML — IEEE HTC Japan 2025 (Published)
- Alpha Thalassemia: A Classification-Based Study — IEEE ASSIC 2025 (Accepted, Oral Presentation)

EDUCATION

B. Tech Bioinformatics — Marwadi University (NAAC A+), Rajkot, Gujarat *2023 – Present*

CGPA: 8.54 / 10 | Semester 7 ongoing

CERTIFICATIONS

- Machine Learning & Python for Data Science — Infosys Springboard (Jan–Mar 2026)
- Data Science Essentials — Cisco Networking Academy (Jul–Sep 2025)
- Complete Data Science, ML, DL & NLP Bootcamp — Udemy (June-Sep 2025)
- Python for Biologists; Bioinformatics Data Processing & Analysis — Marwadi University (2024)